PLECTA: Overlap-aware instance reconstruction of dense filament networks from binary masks

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Abstract

Dense filament networks resist instance segmentation because their two failure modes pull in opposite directions: crossings join distinct objects into one connected component, while gaps in the segmentation divide a single object into several. We present PLECTA, a deterministic method that converts a binary filament-axis mask into overlap-aware two-dimensional instances. It decomposes the skeletonised mask into arms and junctions, alternates exact per-junction continuation matching with globally gated gap matching, and re-estimates tangent, chord, and curvature evidence from the assembled chains, so that a crossing pixel may belong to more than one output instance.

Evaluation used a method-independent partition of each input skeleton into common fragments. On 128 independently seeded held-out scenes from the same procedural generator, PLECTA achieved mean pairwise F_1=0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]), precision 0.868, recall 0.843, recovery 0.799, and adjusted Rand index 0.851, with no failed scenes. Mean F_1 decreased from 0.911 at 20% coverage to 0.775 at 60%, identifying dense crossings as the main remaining regime of error. Against a greedy minimum-turning-angle continuation rule on identical masks, PLECTA gained +0.145 F_1 [+0.129, +0.162] over 50 scenes, and +0.478 over a released implementation run from its own source at a development-selected configuration. Development-only ablations showed that chord-turn evidence, gap bridging, and junction-topology consolidation made the largest contributions.

CNT-sheet scanning electron microscopy is the proof-of-concept application: an upstream synthetic-data-driven CycleGAN/U-Net route prepares the binary masks that let the method run on real micrographs, while SEM width measurement and ribbon rendering are downstream stages that leave the grouping fixed. The evidence supports same-generator synthetic generalisation and a two-field demonstration on real SEM data, not general real-image or three-dimensional reconstruction claims.

Keywords: filament networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

[Igor 1] I must say I'm not quite happy with FilaSeg - we need to come up with a better name of the software. I think we do more than just "Seg", I'd avoid that in the name anyways. I would not go towards something trivial like FIBER or STRING either - not enough distinctive. I find attractive something like REUSS, OCCAM or COSSERAT - with reference to physics/mechanics besides the meaning in abbreviation. But that's your baby of course, your word will be the last.

[Response] Addressed. The method is now PLECTA, after Latin *plecta*, a braid or twisted rope. It carries a meaning tied to what the method does, avoids *Seg*, and is not generic. No expansion is forced onto the letters.

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1 Introduction

[Igor 2] The paper definitely should have one (and only one) focused, bright central idea. My preference - we present a black box that magically transforms microphotographs of entangled bundles of sticky elastic rods into high-resolution, segmented 3D models. We can frame it as the story about CNTs and make it carbon-specific or talk about arbitrary systems (spagetti, straw, bone tissue etc) and make it an image analysis paper. But we certainly should not pick separate elements of the pipeline and discus only those - this way the value of the paper becomes incrementally small.

[Response] Open decision, not yet settled. The single central idea here is *recovering individual filament identity through crossings*, and the paper is built around it end to end: micrograph, mask, instances, geometry. Two constraints pull against the wider framing. The reconstruction is projected and two-dimensional, so *3D models* would overclaim unless we add depth evidence; and the upstream segmenter is a separate contribution with its own evaluation, so folding it in would mean defending two methods at once. The current draft therefore keeps the CNT story as the application and the identity problem as the idea. Please say if you would rather we widen it.

Separating the individual objects in an image of a dense filament network is an instance-segmentation problem with an awkward geometry. The objects are long, thin and open, they touch and cross and occlude one another, and what distinguishes them is local continuation rather than shape or extent. This paper presents a method for that problem. Its proof-of-concept application is scanning electron microscopy (SEM) of carbon-nanotube (CNT) sheets, whose mechanical and electrical response depends on the geometry of the bundle network: length, width, orientation, curvature and contact topology [1–5]. SEM resolves that projected network, but routine image analysis often stops at coverage or orientation distributions [6, 7], whereas instance-resolved analysis requires each visible bundle to remain addressable across gaps, touches and crossings.

That requirement is distinct from semantic segmentation. U-Net and its self-configuring variants can identify filament foreground [8, 9], but a binary foreground mask does not encode object identity: a faded segment splits one filament into several connected components, and a crossing joins different filaments into one. The ambiguity is greatest in dense networks, where several local continuations are geometrically plausible. Compact-object instance priors (bounded blobs, centre-directed flows, star-convex shapes) are poorly matched to long open curves that may overlap [10–12]. Ridge and vesselness filters enhance curvilinear signal but do not assign identity across junctions [13–15], and fibre extraction packages such as FIRE, CT-FIRE, SOAX, FiNTA and DiameterJ report network statistics rather than resolved instances [16–20].

Recovering individual filaments through crossings is itself established prior art, and the design space is well explored. Basu et al. build a minimum spanning tree over centreline particles, delete every bifurcation node, and recombine the segments with a binary integer program over chains of at most three, priced by the integrated squared curvature of a fitted spline; a segment may stay unlinked, and the program is re-solved after refitting until the segment set stops changing [21]. DeFiNe consumes a pre-extracted weighted graph and solves a filament cover as a linear program minimising path roughness, optionally letting several paths cover one edge, so overlaps are representable [22]. SIFNE excises each crossing with its neighbourhood, estimates tip directions towards the local centre of mass, and pairs termini greedily in priority order inside a fan-shaped region under angular and continuity gates, leftover tips becoming ends [23]. De et al. resolve crossovers by partitioning a directed graph [24], and Wegmayr et al. group skeleton branches by lowest mutual angle and learn pixel embeddings for tangled microplastic fibres, sharing junction pixels in principle but assigning them arbitrarily [25]. DNAi detects junctions morphologically, clusters them by single linkage, erases a disk the width of the fibre, and enumerates the pairings at junctions of degree three or four from endpoint angles traced over a short span [26]. GraFT builds a graph from a Frangi-filtered skeleton, derives an angle threshold per component, and traces filaments by constrained depth-first search and a greedy longest-first path cover [27]. Learned alternatives include orientation-aware terminus pairing [28], FibeR-CNN, and recurrent pick-and-trace models [29, 30].

Skeleton graphs, junction excision, continuation angles, distance gates, combinatorial optimisation, iterative refitting and overlapping filament output are therefore all prior art, and none is claimed here. Two distinctions are narrower and survive checking against the sources. First, the methods above either pair endpoints greedily in priority order [23, 25, 27] or enumerate the pairings of a junction only up to degree four [26]; PLECTA instead solves each junction of any degree as one maximum-weight matching in which leaving an arm unmatched is an explicit option at a fixed price, rather than a parity leftover. Second, the methods that optimise globally require intensity evidence or a pre-extracted weighted graph [21, 22], whereas PLECTA reads a binary mask alone. The closest single prior method is that of Liu et al., which likewise works from a mask, reconnects across gaps and permits overlapping output, but pairs termini greedily and depends on a trained orientation network [28].

We introduce PLECTA, a deterministic graph-matching method for dense filament networks, named after the Latin *plecta*, a braid. PLECTA decomposes a skeleton into arms, endpoint stubs, and merged junction nodes. It then alternates exact per-node continuation matching with globally gated gap matching while tangent and curvature estimates expand from local arms to assembled chains. Its output is a set of projected 2-D instance layers in which crossing pixels may be shared.

Reaching real micrographs takes one more component. A complete CNT workflow combines semantic segmentation with algorithmic tracing and transverse intensity profiles [31], and we divide that workflow in two: Pipeline A turns a raw SEM micrograph into a binary filament-axis mask by a synthetic-data-driven CycleGAN/U-Net route [32, 33], and Pipeline B is PLECTA. Without Pipeline A the method could only ever be exercised on masks drawn by hand or emitted by a generator; with the division, grouping can be measured on its own against references whose identities are known exactly, and width rendering cannot move an identity score. Pipeline A carries its own evaluation [33] and is not re-evaluated here. We evaluate PLECTA on 128 unseen scenes from the development generator and on 50 paired clean and degraded scenes, and demonstrate the two pipelines together on two annotated CNT-sheet SEM fields.

2 Materials and methods

2.1 Scope and workflow

The formal input is a binary filament-axis mask $M \in \{0,1\}^{H\times W}$, from which PLECTA reconstructs an overlap-aware collection $\{I_1,\dots,I_N\}$ with $I_n \subseteq M$ and a crossing pixel possibly belonging to several instances. The evaluated grouping stage uses only M; a grayscale SEM image is optional downstream input for width and brightness measurement and cannot revise the fixed grouping.

Figure 1 places the method in the two-pipeline workflow of Section 1: Pipeline A produces the mask from a micrograph, and Pipeline B, which this paper evaluates, is PLECTA. Its five geometric steps read mask geometry alone, and the two matching steps are re-made over eight rounds against frames re-fitted along the chains assembled so far.

What makes the task hard is a single crossing: the mask holds one connected blob from which several arms leave, and nothing local states which arm continues into which. The output is a set of layers rather than a partition, so a pixel inside that blob is counted in both instances that claim it.

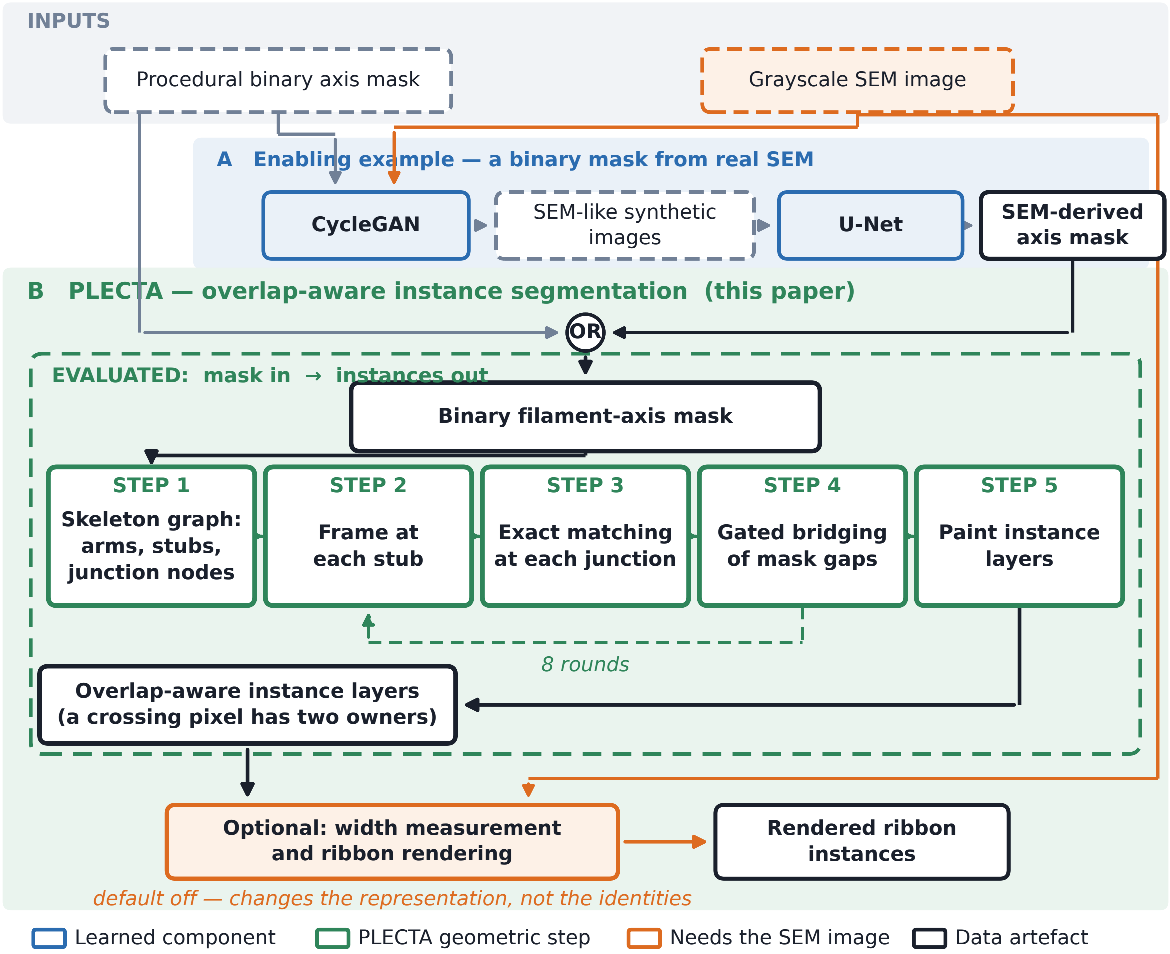

Figure 1. Where PLECTA sits. Pipeline A is the upstream binary segmentation of SEM micrographs; Pipeline B is PLECTA, and the held-out result measures the dashed green box only. The mask entering Pipeline B came from Pipeline A for the SEM fields and from the procedural generator in every synthetic experiment. Colours: blue, a learned component; green, a PLECTA geometric step; orange, a dependency on the grayscale image; dark, a data artefact.

2.2 Synthetic data and evaluation split

Procedural scenes preserve the ordered centreline $\mathcal{C}_k$ and rendered support Ω_k of every filament k; supports remain separate, so several filaments may own the same crossing pixel. Besides the exact overlap-aware reference, the generator provides an undegraded one-pixel axis mask, a degraded thin mask containing small gaps and irregularities, a combined thick foreground, and an SEM-like image. The degraded one-pixel axis is the input evaluated throughout. Areal coverage is the union area of the thick supports divided by image area, not the occupancy of the thin input mask, and Section 3.2 shows it is a proxy for difficulty rather than a cause of it.

Development used 84 scenes spanning 20–60% coverage. The held-out set comprised 128 different seeds, 32 scenes at each of 20, 30, 40, and 60% coverage: an independent sample from the same generator, not an independent distribution. A separate 125-scene locked set remains unevaluated. The compact seed manifest records every seed block and an executable verifier checks counts, disjointness, and available scene metadata.

A separate controlled robustness set of 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage, was scored once from each geometry's clean axis and once from its degraded mask, after the configuration was fixed.

2.3 CNT SEM data and mask sources

Two manually annotated CNT-sheet SEM fields were available for exploratory analysis: B58_100 (1024×1536 px, 280 filaments) and B58_110 (1024×1536 px, 430 filaments). The nominal horizontal field width is 1.66 µm; project metadata associate it with 1536 px, or approximately 1.08 nm px^{-1}. The annotation is one human interpretation of projected, locally ambiguous crossings, and both fields informed development.

Two inputs were derived per field. The first was the skeleton of the manual instance annotation. The second came from Pipeline A: an nnU-Net trained on paired synthetic images and masks, with CycleGAN translation as the route from procedural masks to SEM-like training images [32, 33]. Together the two conditions separate what grouping can do from what the mask allows it to do. B58_100 appeared in that model's training data, whereas B58_110 was verified absent from the relevant training set, so only B58_110 provides a training-clean observation of mask transfer, and $n = 2$ precludes inference.

[Needed before submission: A fuller account of Pipeline A belongs here, to be written by the author: the CycleGAN configuration used to translate procedural masks into SEM-like images, the paired synthetic corpus it produced, the nnU-Net variant and training split trained on that corpus, the optimisation settings, and how the predicted masks were selected and post-processed before entering PLECTA. Take these from the model archive rather than reconstructing them.]

Mask quality was compared after skeletonisation with a symmetric 3-px tolerance, which avoids rewarding mask thickness, and reconstruction scores were computed within each mask's own visible common-fragment population.

[Needed before submission: Author metadata: SEM instrument and detector, accelerating voltage, working distance, specimen preparation, CNT material identity, annotation protocol, and confirmation of the 1536-pixel calibration against the stored image dimensions.]

2.4 PLECTA reconstruction

2.4.1 Skeleton graph and local frames

PLECTA binarises M at > 0, skeletonises it, and iteratively removes spurs no longer than 3 px. Skeleton pixels with degree at least three form junction clusters, and adjacent clusters are merged into nodes. Connectors up to 5 px are absorbed into one node; connectors up to 14 px merge nearby nodes but remain real arms; free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm with one stub at each end.

A stub frame contains its tip, outward tangent, and signed curvature, estimated by an arclength polynomial fit and considered reliable only with at least three samples spanning at least 3 px. Local frames use 24 px of support; once links form chains, the support expands to 55 px, so that a stub too short to have a direction of its own borrows one from the chain it has joined.

The cost of linking stubs a and b combines three angular terms, curvature continuity and, for gaps, a length penalty (Figure 2):

$$C_{ab} = w_d\,\theta + w_t\,q(d)\,(\varphi_a + \varphi_b) + w_\ell\,\frac{d}{\ell_0} + 30\,w_\kappa\,|\kappa_a + \kappa_b|, \qquad q(d) = \min\Big(1, \tfrac{d}{d_0}\Big), \tag{1}$$

with d the tip separation, θ the reversal angle between the two outward tangents, and φ_a, φ_b the turns from each tangent onto the tip-to-tip chord, measured at their own vertices. Junction links set $w_\ell = 0$, a junction having no gap to pay for. The reversal angle ignores the chord, which keeps it meaningful when the tips are three pixels apart and the chord direction is quantisation noise;

the chord turns instead see lateral offset, so two parallel arms that never meet are penalised. Because it is the chord terms that degrade over short separations, the fade $q(d)$ withdraws them as the tips approach rather than letting an unstable angle dominate. The constant 30 places curvature, in px^{-1}, on the scale of the angular terms; the weights, ℓ_0 and d_0 are fixed in the public configuration.

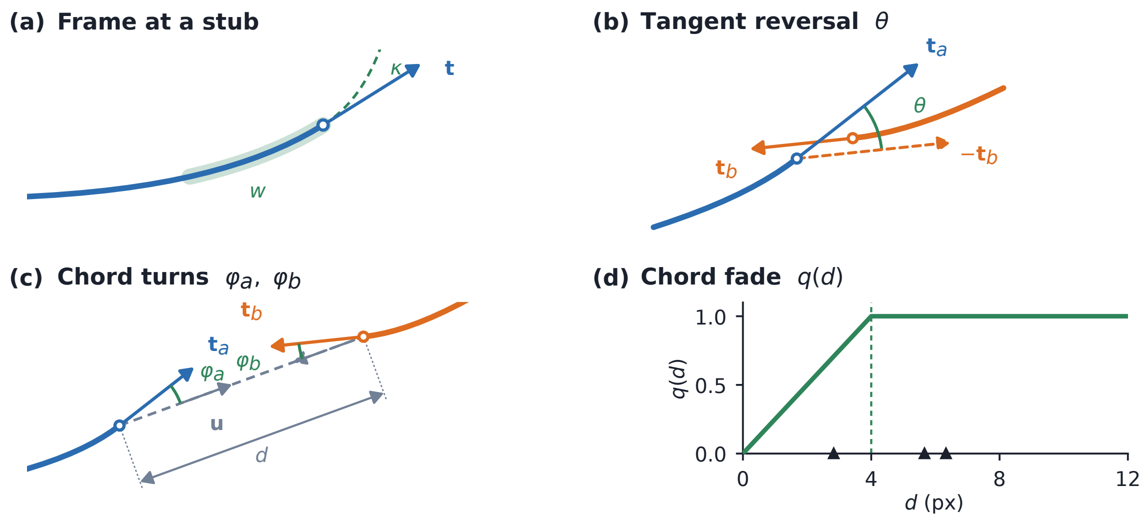

Figure 2. The stub frame and the terms of the pairing cost. (a) The least-squares arclength fit over window w, read outward from the tip at $s = 0$, giving outward tangent $\mathbf{t}$ and signed curvature κ; the fit is quadratic when the span is at least 18 px and linear otherwise. (b) $\theta = \arccos(\mathbf{t}_a \cdot -\mathbf{t}_b)$, drawn where the reversed tangent lies and again from the common origin where the angle is read. (c) The chord turns φ_a, φ_b onto the chord $\mathbf{u}$ joining the tips. (d) The fade $q(d)$ against tip separation; triangles mark the six real separations at the crossing of Figure 4.

2.4.2 Candidate gates and exact matching

Leaving a stub unmatched is priced rather than forbidden, and the price p_i also fixes admissibility: an edge is offered only when $C_{ij} < p_i + p_j$. Exact maximum-weight partial matching then maximises $\sum(p_i + p_j - C_{ij})$ over disjoint pairs, which minimises the total link cost plus p_i for every stub left free. At full strictness junction stubs carry $p_i = 0.62$, so a junction edge needs $C < 1.24$; gap edges need $C < 0.60$ and must in addition satisfy $d \leq 85$ px, $\theta \leq 0.40$ rad and $\varphi \leq 0.28$ rad at each end. Gap gates are the tighter set because a gap link asserts a continuation across evidence that is missing rather than merely ambiguous; the values themselves were fixed on development scenes, and the complete set is recorded in the public configuration. Rounds $r = 0, \ldots, 7$ apply strictness $\alpha_r = 0.85 + 0.15\, r/7$, scaling the unmatched prices and the two gap angle limits and nothing else, so early rounds commit only to pairings that survive a loose fallback; the eighth round, $\alpha = 1$, is the only fully strict one and is the one returned. In full:

1. Skeletonise M, prune spurs, merge junction clusters into nodes, and store the remaining paths as arms with a stub at each end.
2. For $r = 0, \ldots, 7$, re-solving both matchings from scratch and carrying only the chain structure forward from round $r - 1$:
   (a) Re-fit every stub frame along the chain it now belongs to.
   (b) At each node, form candidate edges between stubs of different arms with finite frames and cost, keep those admissible at α_r, and solve one maximum-weight matching per node.
   (c) From the still-unmatched reliable stubs, form gap candidates on different arms and different nodes that pass the distance gate and the α_r-scaled angle gates, and solve one global maximum-weight matching over the admissible ones.

3. Return the links of round $r = 7$.

Figure 3 enumerates the outcomes this admits: every stub is either paired or pays the unmatched price, and there is no third possibility. The priced option is what lets a filament genuinely end, and what stops a T-junction from inventing a continuation for the arm that has no partner.

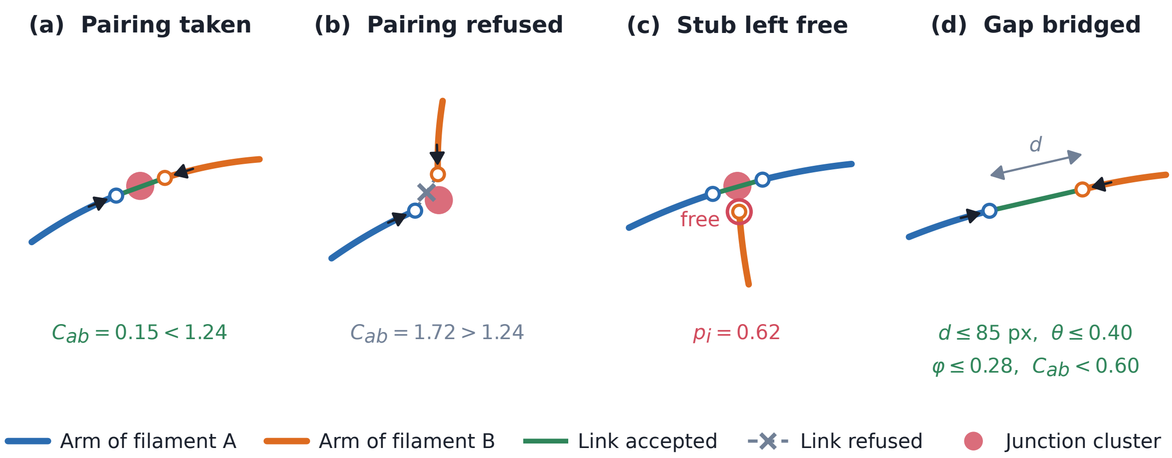

Figure 3. Every decision the linker can make. (a) A junction pairing accepted below the admissibility limit $2p_i$. (b) A competing pairing at the same node refused, its cost exceeding $2p_i = 1.24$. (c) A stub left unmatched because paying $p_i = 0.62$ beats any edge it has. (d) A gap joined between two stubs the junction stage left free, under four gates at once. Idealised drawing; measured values are in Figures 2 and 4.

Solving a junction as one matching rather than one edge at a time is the distinction claimed in Section 1, and Figure 4 shows what it buys on real nodes. A locally attractive edge can be declined, as at the three-arm node of Figure 4e. Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense: over 20 development scenes they never disagreed at nodes of degree 2–4 ($n = 542$), disagreed at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$), where the joint solution was better by a median of 0.90 in total cost. That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, a cheap early choice propagates and the difference is no longer confined to dense nodes (Section 3.3).

2.4.3 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the fixed JSON configuration.

2.5 Optional SEM characterisation and ribbon rendering

Two optional stages act after grouping, both off by default and both governed by limits recorded in the fixed configuration. Perpendicular SEM profiles, sampled along each ordered chain away from junctions, attach apparent width, brightness, uncertainty, and a quality flag to an existing instance without altering its arm membership. Ribbon rendering then interpolates accepted gaps, resamples and smooths the centreline, and draws the chain at its fitted width, excluding junction blobs. Silhouette absorption, which would union rendered instances, has threshold zero

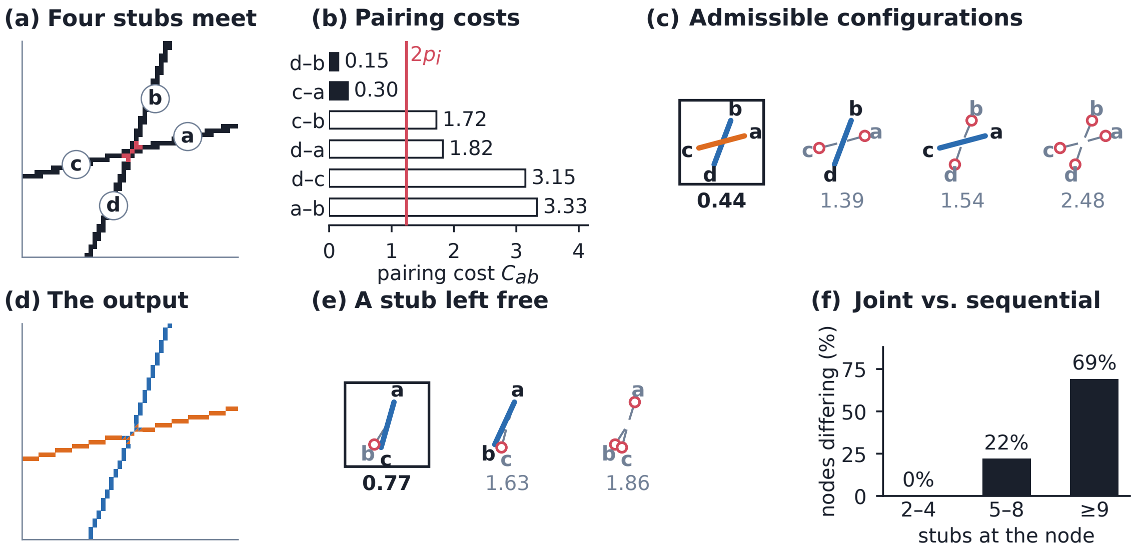

Figure 4. The pairing at a crossing, solved jointly and exactly; every number shown is measured. (a) A four-arm crossing (development scene `cov20/synth_0001`, node 9), arms named a to d anticlockwise by outward tangent, junction pixels red. (b) All six pairwise costs; open bars are the four refused by $C_{ab} < 2p_i = 1.24$. (c) Every admissible configuration, scored jointly; the boxed one is returned. (d) The result: two instances sharing the seven junction pixels. (e) A three-arm node (`cov40/synth_0001`) at which an admissible edge costing 1.01 is declined: taking it scores 1.63 overall, against 0.77 for pairing the other two arms and paying to leave the third free. (f) Joint against sequential cheapest-edge-first solutions, by node degree, over 20 development scenes.

by default; neither it nor any post-hoc continuation merge is part of the evaluated algorithm or contributed to the held-out result.

2.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1 (Figure 5). For each scene the visible input skeleton is partitioned once into method-independent elementary fragments, and each fragment receives a procedural reference identity and a PLECTA instance identity. Because the partition is built from the input mask alone, every method compared is cut into the same fragments. A pair of fragments is a true positive when it shares both identities, a false positive when it joins different reference filaments, and a false negative when it separates fragments of one. Thus

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}. \tag{2}$$

This is a clustering metric related to the Rand family [34, 35]. It scores grouping rather than rendered width, and weights filaments with many fragments more strongly because their pair count is quadratic, so we also report adjusted Rand index [36], variation of information split and merge components [37], and the fraction of reference filaments recovered by one dominant predicted instance at 0.8 purity and coverage.

Held-out means and density strata are accompanied by 95% nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810), resampled with stratification by density; the clean-minus-degraded analysis resamples paired scene differences within density. Active-component ablations are one-at-a-time development experiments on all 84 development scenes and do not reuse the held-out set.

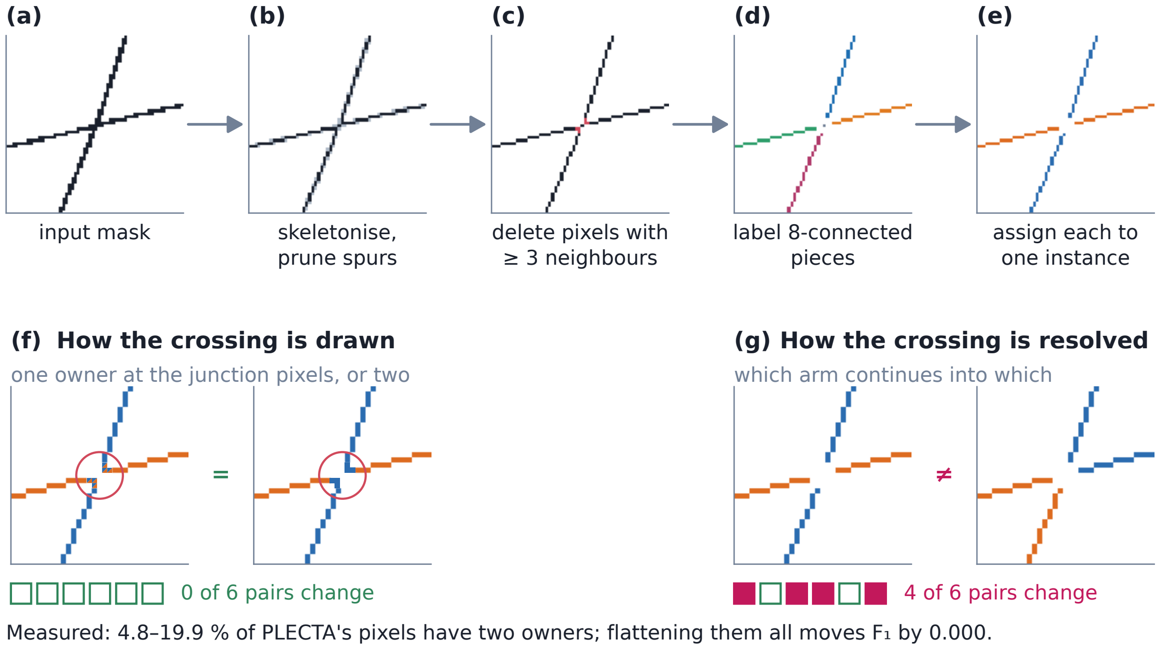

Figure 5. The common-fragment metric, on a real crossing. **(a)**–**(e)** fragments are built from the input mask alone: the mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded. Each is then assigned to one reference and one predicted instance by majority overlap, and the score is counted over fragment *pairs*. **(f), (g)** the consequence: no fragment contains a junction pixel, so redrawing the crossing, giving its pixels one owner instead of two, leaves all six fragment pairs untouched, while resolving it differently changes four of the six. The metric therefore scores how a crossing is resolved and not how it is drawn.

3 Results

3.1 Held-out synthetic reconstruction

PLECTA completed all 128 held-out scenes with 0 failures. Mean common-fragment pairwise F_1 was 0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]); Table 1 gives precision, recall, reference-instance recovery, adjusted Rand index and both variation-of-information components, overall and by density. These scenes used unseen seeds from the same procedural generator as development data, so this is same-generator generalisation and not distribution transfer (Section 4.2).

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information, for which lower is better; higher is better for the other metrics.

Group	n	F_1	Precision	Recall	Recovery	ARI	VI split	VI merge
All	128	0.854	0.868	0.843	0.799	0.851	0.256	0.216
20%	32	0.911	0.928	0.896	0.918	0.907	0.137	0.093
30%	32	0.874	0.892	0.858	0.828	0.870	0.211	0.161
40%	32	0.858	0.864	0.853	0.796	0.855	0.242	0.224
60%	32	0.775	0.787	0.765	0.656	0.772	0.436	0.388

Performance decreased with scene density: mean F_1 fell from 0.911 [0.888, 0.933] at 20% coverage to 0.775 [0.760, 0.790] at 60%, monotonically through the intermediate strata. The same decline appears on the paired robustness set (Figure 6): the two curves separate at low coverage and meet at 50–60%, degradation costing least exactly where the score is lowest, because there the tangles rather than the gaps set the difficulty. Per-scene runtime had mean 2.54 s, median 1.12 s, 90th percentile 6.42 s, and maximum 34.06 s; these timings are hardware-specific.

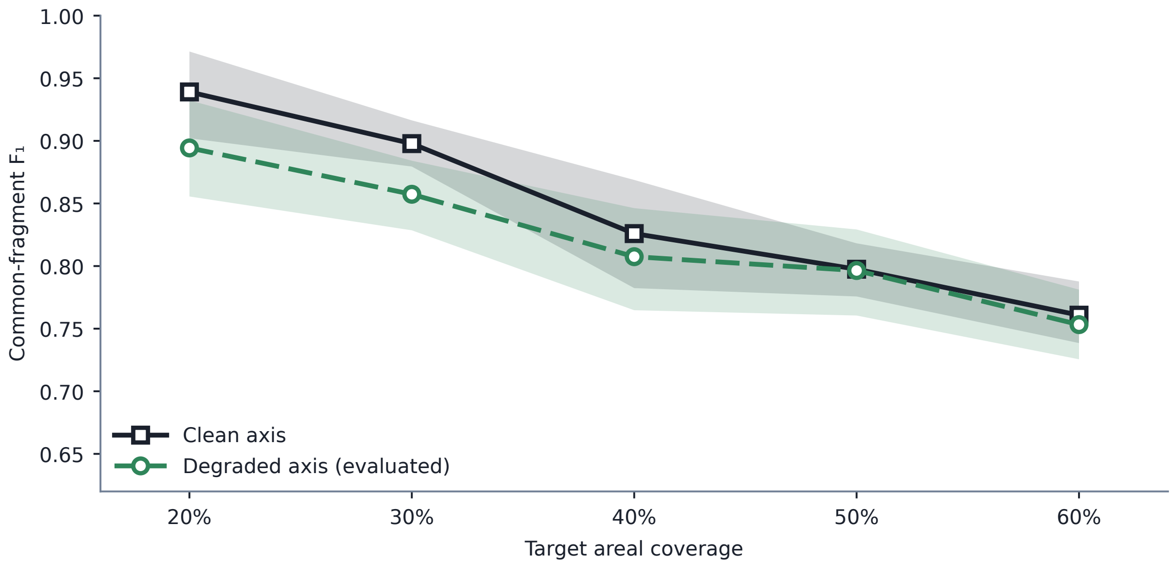

Figure 6. Coverage response on the 50 paired scenes, ten per coverage level, each scored twice: from the degraded 1 px axis mask that the paper evaluates, and from the generator's undegraded axis. Lines are means of per-scene common-fragment F_1; bands are 95 % percentile bootstrap intervals for the mean, resampled within coverage stratum (20 000 replicates, seed 20260811). Target areal coverage is the generator setting used to dial in centreline density.

3.2 Which density axis matters

Coverage is the fraction of the frame filled by the rendered bundles, so it rises both when filaments are added and when the same filaments are drawn thicker, and only the first changes the grouping problem. A factorial separates them: twelve geometries, three centreline length densities crossed with four seeds, were each rendered at bundle widths of 6, 11 and 16 px. Within one geometry the input axis mask and the overlap-aware centreline reference are byte-identical across the three widths, verified by SHA-256, while the union of the thick supports changes: areal coverage varies by up to a factor of 2.49. Any movement in the score across those three renderings would mean that the method or the metric reads the silhouette.

Across all 12 geometries the within-geometry F_1 range was 0.0000, that is, exactly zero in 12 of 12 (Table 2). Areal coverage therefore cannot causally affect the grouping score; PLECTA reads the axis mask, and the metric scores centreline fragments, so neither sees the width the bundles were drawn at. Centreline length density does move it, mean F_1 falling from 0.920 to 0.836 across the three densities of Table 2. The coverage bands of the three length densities also overlap, so coverage does not identify a difficulty level on its own. Density strata reported by target areal coverage should therefore be read as a proxy for the length density accompanying them under this generator.

3.3 Comparison with prior linkers

Three comparators were run on identical input masks with one scorer: a greedy minimum-turning-angle continuation rule, which tests the paper's central claim directly, and two published methods, DNAi [26] and GraFT [27], each executed from its own released source. All were scored through the same conversion and metric code as PLECTA, and that path was gated on parity:

Table 2. Bundle width against centreline length density, 36 development scenes. Within each row the three bundle widths share one byte-identical axis mask and one reference, so the spread in areal coverage carries no score difference. Length density is centreline pixels per image pixel.

Centreline length density	n	Areal coverage	Bundle width	F_1
0.0193	12	0.118–0.309	6, 11, 16 px	0.920
0.0372	12	0.218–0.505	6, 11, 16 px	0.878
0.0548	12	0.305–0.644	6, 11, 16 px	0.836

before any comparator number was believed it had to reproduce PLECTA's stored F_1, adjusted Rand index and both variation-of-information components, because a faulty conversion yields plausible rather than obviously wrong scores. Each comparator received a parameter translation selected on the same 84 development scenes PLECTA was configured on, and every adaptation below was checked and found to favour the comparator.

Greedy continuation. The contribution claimed in Section 1 is that a junction is better solved as one exact matching, with an explicit priced option to leave an arm unmatched, than by pairing endpoints greedily in priority order, the rule used by SIFNE, by GraFT's path cover, and by the continuity baseline of Wegmayr et al. [23, 25, 27]. The comparator therefore skeletonises the same mask, prunes spurs, splits the skeleton into branches, and greedily accepts the lowest-turning-angle pairing between branch ends within a distance and a turning-angle gate, both gates fixed by a grid sweep for best mean development F_1. On 50 scenes generated after both configurations were fixed, PLECTA scored 0.822 against 0.676 and was better in 49 of them; Table 3 gives that difference and every other, the field-standard measures ranking the two as the in-house one does. On clean axes the difference was +0.128 [+0.113, +0.144], so it is not an artefact of mask degradation.

Table 3. All four methods on the degraded masks, identical input masks and one scorer. $n = 50$ scenes and 3638 reference crossings, except GraFT, over the 47 it completed. Variation of information is in bits and lower is better; higher is better for the rest. The last two rows are defined in Section 3.4. The final column is the paired difference against the greedy continuation, density-stratified 95% bootstrap; the other paired differences are in the text.

Measure	PLECTA	Greedy	DNAi	GraFT	PLECTA − greedy
Common-fragment F_1	0.822	0.676	0.343	0.490	+0.145 [+0.129, +0.162]
Precision	0.831	0.785	0.462	0.666	+0.046 [+0.029, +0.062]
Recall	0.814	0.599	0.284	0.400	+0.215 [+0.196, +0.235]
Reference-instance recovery	0.765	0.542	0.252	0.304	+0.223 [+0.201, +0.245]
Adjusted Rand index	0.817	0.670	0.331	0.479	+0.148 [+0.131, +0.165]
Variation of information ↓	0.573	1.036	2.107	1.558	-0.463 [-0.509, -0.418]
Crossings resolved exactly	0.809	0.729	0.581	0.683	+0.065 [+0.047, +0.083]
Resolution index	0.778	0.736	0.401	0.639	—
Median s per scene, 20/60%	0.14–4.1	—	0.04–0.6	2.44–29.7	—

Exact against greedy matching, matcher only. That baseline replaces the whole linker; a second arm isolates the claim, keeping the graph, cost, admissibility gate and round schedule and replacing only the per-junction maximum-weight matching by greedy cheapest-first acceptance. Exactness was worth +0.080 F_1 [+0.070, +0.089] over 30 development scenes, on 29 of 30 of them, and +0.090 [+0.080, +0.100] on clean axes; the failure is division, not merging, with variation-of-information split rising 0.252 to 0.500 at unchanged merge, 30 % more instances emitted and crossings resolved exactly falling 0.861 to 0.668. The relative loss grows monotonically with degree, -21 % at degree three to -35 % at degree six, but in absolute terms peaks at degree four

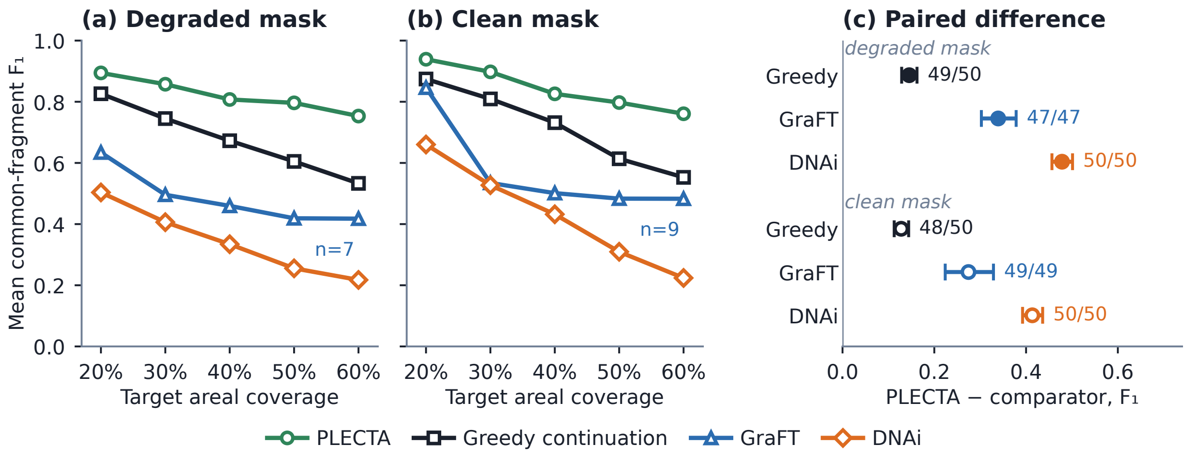

Figure 7. The comparison of Table 3 by coverage stratum. **(a)** degraded and **(b)** clean input mask: mean common-fragment F_1, $n = 10$ scenes per point except where labelled otherwise; plain means, no interval is drawn and none is claimed. GraFT exceeded its per-scene wall-clock budget on 3 of the 50 degraded scenes and 1 of the clean ones, the hardest of the densest stratum, so its curve is optimistic. **(c)** aggregate paired difference with its density-stratified bootstrap 95 % interval (20 000 replicates), filled markers degraded and open markers clean; the fraction beside each interval is the number of scenes on which PLECTA scored higher, of those both completed.

(-0.252), where most crossings lie; and the greedy arm runs at parameters selected with the exact matcher in the loop, a bias in that matcher's favour.

DNAi. DNAi post-processes a binary segmentation by the sequence described in Section 1; its segmentation network was bypassed and only that stage used, so grouping is not confounded with segmentation quality. At its released defaults it scored F_1=0.195 on the degraded masks, which is not a fair measure of it: its single-linkage clustering of junction pixels at 10 px chains together crossings that lie closer than that, and at 60% coverage the largest cluster spans 89 px and is replaced by one small erasure disk, leaving the crossings along it unseparated and the whole tangle a single instance. Reducing that distance to 2 px on development scenes raised development F_1 from 0.211 to 0.378 and cut variation-of-information merge from 2.07 to 0.66, confirming the mechanism; dilating the input to a thicker mask was also tested across 41 development configurations and made DNAi worse at every setting, so it was not used. At its best development-selected configuration DNAi scored 0.343, a paired difference of +0.478 [+0.456, +0.501] over 50 scenes, and +0.414 [+0.392, +0.436] on clean axes, the gap widening monotonically with density (Figure 7). Both methods must be read against the one-instance-per-connected-component floor at their own density, since that floor is itself strongly density-dependent and a pooled value would flatter any method at 60% coverage.

That margin is a limitation of domain and density, not a general ranking. DNAi is built for DNA-fibre micrographs, which are sparse: its own bundled test image contains five junction clusters, whereas our scenes contain between 23 and 183, and where the network is sparse and the axis clean it recovers 0.660 at 20% coverage. Two structural differences explain the rest. It commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can; and it resolves only junctions of degree at most four, measurably so, resolving 0.016 of the crossings above that degree. Those are about a tenth of the crossings but carry 37.2 % of same-filament fragment pairs on the degraded masks, one such junction severing every pair that spans it.

Restricting the score to the evidence a degree-limited, gap-blind method can reach separates capability from judgement. On clean axes the difference falls from +0.414 to +0.284 [+0.262,

+0.307], still favouring PLECTA in 49 of 50 scenes and at every coverage level. DNAi gains far more from the restriction than PLECTA does, which is what shows the restriction genuinely removes its handicap. So roughly 31 % of the margin is coverage of cases the comparator cannot attempt and 69 % is the outcome of decisions both methods make. Consistent with that, after its clustering is repaired DNAi's remaining errors are dominated by division, a third of them at junctions of degree four, which it does attempt.

GraFT. GraFT builds a graph on a skeleton, derives an angle threshold per component, and traces filaments by constrained depth-first search followed by a greedy longest-first path cover. Its published entry point begins with a Frangi and Otsu front end for grayscale ridge images, which has nothing to do on a two-valued axis, so the mask was injected at its skeleton-to-graph stage, everything downstream being its own code unchanged. That substitution favours GraFT: run end to end on the grayscale images it scores 0.100, near the connectivity floor, because its segmentation stage recovers only a third to a half of the reference skeleton. Its minimum continuation angle was re-tuned on development scenes, gaining it a further 0.04. GraFT scored 0.490, a paired difference of +0.339 [+0.303, +0.378] favouring PLECTA in every one of the 47 scenes both methods completed, and +0.275 [+0.224, +0.329] on clean axes. Its errors are divisions rather than merges (variation of information split 1.166 against merge 0.392), which follows from a greedy cover that cannot revise an early commitment once the scene collapses into one connected graph, as it does above 30% coverage. 3 of the densest degraded scenes did not finish inside a 45-minute budget and are excluded from its means; they are the hardest scenes of the set, so that exclusion flatters GraFT. Neither released method establishes a general ranking, and Section 4.2 states what these comparisons leave untested.

Per-scene cost was measured warm, single-threaded and pinned to one core, timing each methodś own computation only (Table 3; 3 GraFT scenes exceeded 600 s and are censored). These are independently optimised codebases, so absolute times measure implementation as much as algorithm; the transferable quantity is scaling, and the log–log exponents against foreground pixels are 2.11, 2.65 and at least 2.01. PLECTA grows fastest, but only GraFT is heavy-tailed.

3.4 Crossing resolution and crossing representation

Two distinct things are meant by an overlap-aware result, and the metric treats them differently. The first is whether a crossing is *resolved* correctly: of the arms meeting at a junction, which continues into which. That is the difficult decision, and it is what the common-fragment metric measures (Figure 5), so every score in this paper rewards getting it right. The last two rows of Table 3 measure it directly: over the 3638 reference crossings of the degraded set, the fraction whose incident arms a method partitions exactly as the reference does. It orders the methods as the grouping scores do, PLECTA's paired margins over the greedy baseline, GraFT and DNAi being +0.065 [+0.047, +0.083], +0.123 [+0.085, +0.165] and +0.185 [+0.158, +0.211]; clean axes order them the same way, PLECTA at 0.827. Chance is high here, since leaving every arm unpaired already resolves 0.709 of crossings, so the chance-corrected resolution index accompanies the raw rate in the table and never replaces it. The measure shares its assignment thresholds with the common-fragment metric, so it is a sharper view of the same evidence rather than independent confirmation of it.

The second is whether the emitted raster *represents* a crossing pixel as belonging to both filaments. That is a property of the output format, not of the grouping decision, and the metric cannot see it. Flattening either method's layers to a single label image accordingly changes F_1 by at most 0.0000 for PLECTA and 0.002 for GraFT.

The representation is instead verifiable directly: on the comparison set, reference instances share 11 % of their pixels with another instance at 20% coverage, rising to 39 % at 60%. PLECTA emits 5–20 % shared pixels; GraFT, whose output is close to a hard partition, about 2 %. Which pixels those are is less flattering: PLECTA recovers 0.463 of the pixels the reference gives to more than one instance, the only method here that recovers most of them, at a precision of 0.134, marking 3.6–4.0 times more shared pixels than the reference contains, because a chain is painted with every pixel of every junction cluster it touches — a property of the rendering, not of the grouping. Neither that output nor GraFT's near-hard partition, a real structural limitation, is anything the F_1 rewards or penalises; Section 4.2 takes up what that means for the comparison.

Figure 8 shows what the score corresponds to scene by scene, taking at three coverage levels the scene whose F_1 is closest to the median, so the rows are typical rather than selected. Its fourth column adds the fitted widths the full pipeline delivers, included because the comparison with the reference is only legible once instances are drawn at width; rendering redraws the instances the grouping produced and cannot merge or divide them.

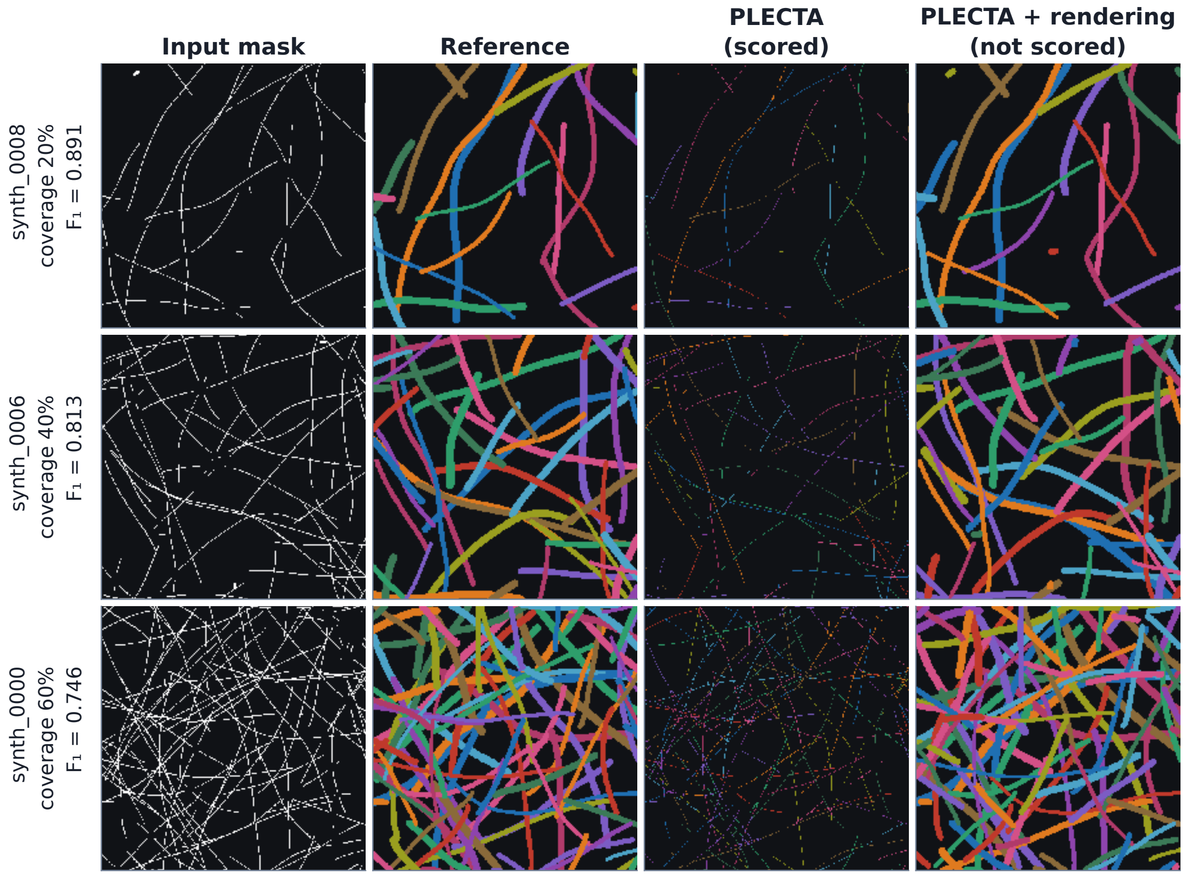

Figure 8. Reconstruction at three coverage levels, each the scene closest to the median F_1 of its stratum. Columns: the degraded input mask, the overlap-aware reference, the evaluated centreline instances, and the same instances after optional width rendering. Only the third column is scored. Instance colours are arbitrary and not matched between columns.

3.5 Input-mask robustness

On the separate 50-scene controlled set, mean F_1 was 0.822 from degraded axes and 0.844 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113, 0.0333]). Modelled gaps and raster irregularities therefore caused a small resolved loss under this generator. Mask thickness costs more (Figure 9), which matters because it is the upstream segmenter, not the method, that fixes that thickness. Each width variant is scored against common fragments rebuilt from its own input mask, so the question is how well the method groups the skeleton it is handed; the decline with width is a thicker mask skeletonising into a messier graph, not a change in the grouping problem.

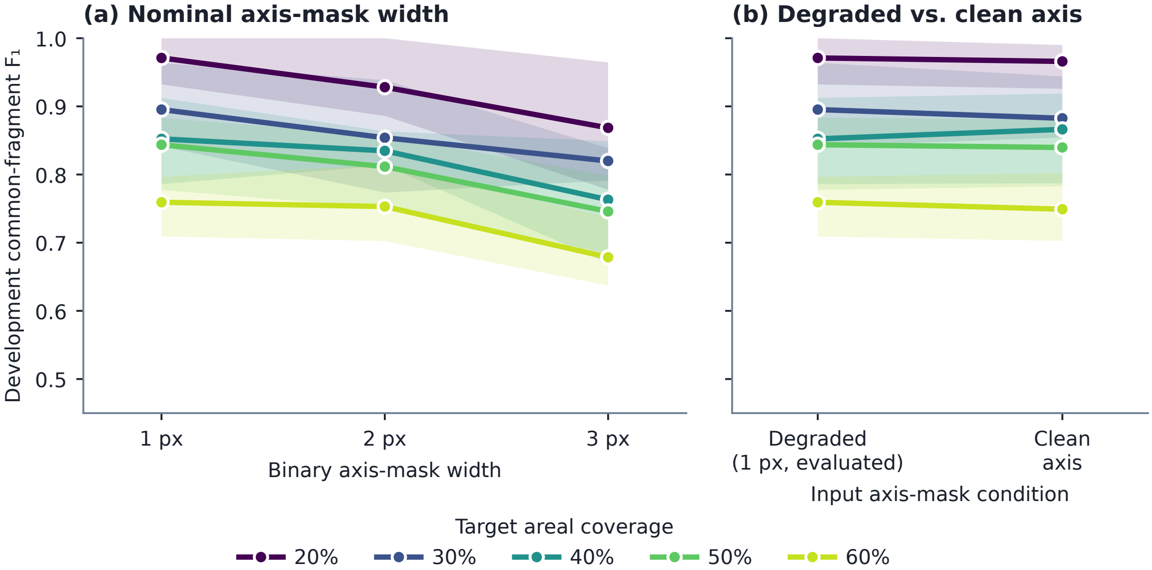

Figure 9. How PLECTA responds to the input mask it is handed, on development scenes. (a) Nominal binary axis-mask width. (b) The degraded 1 px axis against the generator's undegraded axis. Lines are means over the four development scenes at each coverage level; bands are the min–max range, a description of spread and not an interval. Exploratory, $n = 4$ scenes per point, no inference.

3.6 Development ablations and downstream rendering

One-at-a-time ablations were evaluated only on the 84 development scenes (Table 4). The three largest losses relative to the evaluated configuration were, in order, the junction chord-turn term (-0.1858), gap matching (-0.1394) and connector-based junction-cluster consolidation (-0.0634); every other component, chain-extended frames and the round schedule included, mattered less. These are development values, not held-out effect estimates.

Table 4. One-at-a-time component ablations on the 84-scene development set. Negative ΔF_1 values denote losses relative to the evaluated configuration.

Development variant	F_1	ΔF_1	ARI
Fixed configuration	0.867	+0.000	0.864
No Gap Bridging	0.728	-0.139	0.722
Single Round	0.847	-0.020	0.844
No Chain-Extended Frames	0.846	-0.021	0.842
No Annealing	0.863	-0.004	0.860
No Curvature Term	0.863	-0.005	0.859
No Join_Px	0.845	-0.022	0.841
No Junction-Cluster Consolidation	0.804	-0.063	0.799
No Junction Chord-Turn	0.681	-0.186	0.675

The core output is a grouping, not a set of connected curves: accepted gap links establish identity but are not painted, so on the development set 0.653 of core instances arrive as more than one disjoint pixel run and 0.663 contain a branch point. Any downstream step that assumes one connected curve per instance must account for this. Default Stage 4 ribbon rendering scored development F_1=0.860, a change of -0.0068, and reduced those fractions to 0.000 and 0.005, so rendering buys connectivity at a small cost in the identity score without changing chain membership. Enabling the separate 0.6 silhouette-absorption rule reduced F_1 further to 0.857, and because absorption can merge identities it remained disabled. None of these topology measurements validates apparent width.

The negative control of Section 3.2 is worth seeing scene by scene (Figure 10), because a mean can hide a null. PLECTA's errors are also two-sided, where a fragment F_1 can be reached one-sidedly: it sits on the precision–recall diagonal, whereas the connected-component floor recovers 0.850 of same-filament pairs at a precision of only 0.047.

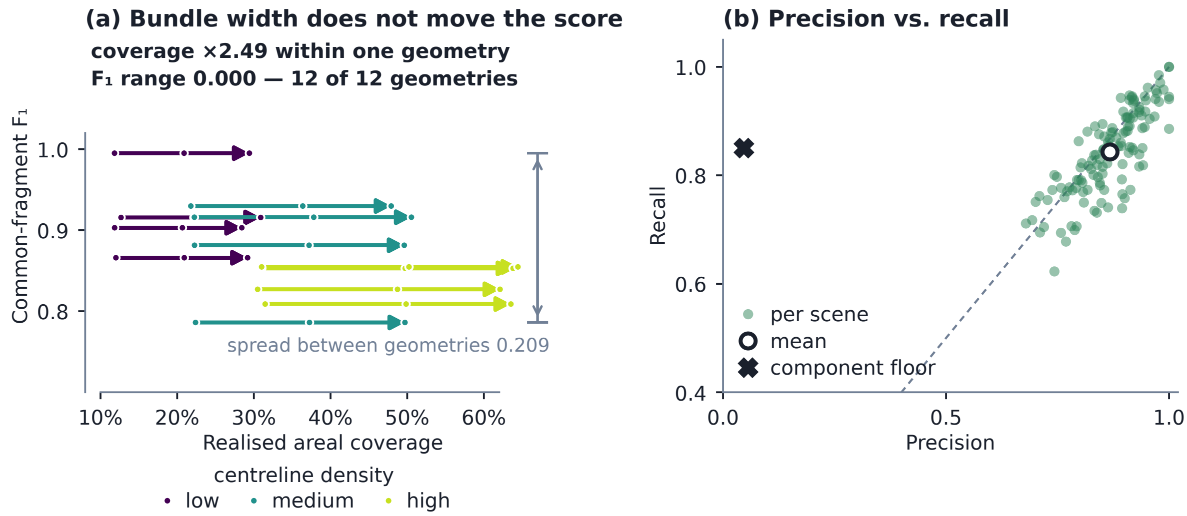

Figure 10. The width factorial, scene by scene. Each arrow is one of the 12 geometries, drawn from its narrowest to its widest rendering. Along an arrow the centreline geometry and the input mask are fixed and only the bundle width changes, so realised areal coverage moves by up to a factor of 2.49 while F_1 moves by 0.0000. The bracket marks the spread between geometries, the scale against which that null should be read.

3.7 Two-field CNT SEM mask-source case

The two real SEM fields are the proof of concept for the whole route, and are exploratory development cases. From the manual-derived axis, an oracle-input condition, PLECTA reached F_1=0.882 on B58_100 and 0.837 on B58_110; the U-Net condition, and the agreement of those axes with the manual skeleton, are in Table 5. The distance between the conditions is the practical dependence on upstream mask quality. These values are descriptive and do not estimate real-image generalisation, for the reasons set out in Section 4.2.

Table 5. Exploratory reconstruction from manual-derived and U-Net axes in two CNT SEM fields. Axis recall and precision compare the U-Net skeleton with the manual-derived skeleton using a symmetric 3-px tolerance.

Field	Input mask	Axis recall	Axis precision	F_1	ARI
B58_100	Manual-derived	–	–	0.882	0.882
B58_100	U-Net	0.782	0.753	0.438	0.435
B58_110	Manual-derived	–	–	0.837	0.836
B58_110	U-Net	0.729	0.887	0.452	0.449

4 Discussion

PLECTA addresses a narrow problem: assigning fragments of a binary filament-axis mask to overlapping projected instances. Explicit geometry recovers substantial identity there with no instance-labelled training data and no SEM intensity, and what recovers it is the combination of local and accumulated evidence, neither of which suffices alone: a junction is resolved by mutual compatibility rather than by the cheapest available continuation, and the evidence for that judgement is re-read along the chains as they assemble. The ablations locate that effect without proving each term independently necessary: the two large losses correspond to the two kinds of discontinuity, the junction and the missing mask segment, while accumulated context is worth distinctly less than either.

4.1 Density, masks, and optional rendering

Difficulty follows the density of centrelines rather than the amount of ink, so what costs the score is the number of locally plausible alternatives at a crossing. The rest of the sensitivity belongs to the upstream segmenter, which fixes the thickness of the mask and therefore how clean a skeleton PLECTA is given. PLECTA can bridge a short omission, but it cannot infer a filament absent from the mask, nor reliably reject a false axis introduced upstream. Supplying identical masks isolates grouping in an algorithmic comparison; it does not establish end-to-end SEM performance, which is why the two-field case is a demonstration of the joined pipelines rather than a measurement of them.

Ribbon rendering, being downstream of grouping, buys connectivity rather than identity, because the core paints arms and junction clusters but not the gap links that carry identity across a break. Silhouette absorption differs in kind, since it changes the partition itself, and so stays optional; width and brightness can be reported for characterised instances, but their physical accuracy requires separate references.

4.2 Evidence boundary

The principal limitation is distributional. The held-out scenes use independent seeds but the same procedural generator family used for development. Their confidence intervals quantify sampling variation within that family, not transfer to another simulator, specimen, microscope, or segmentation pipeline. The ablation and rendering studies are development-only and remain secondary to the fixed held-out result.

The real SEM study is deliberately a case study with two fields, not a population validation. Manual-derived masks are an oracle-like input condition: they test grouping when the axis evidence is curated, not automatic SEM-to-instance performance. B58_100 is training-contaminated for the U-Net condition and B58_110 documented as training-clean, so only the latter illustrates an unseen upstream mask, and one field cannot estimate generalisation. The two mask sources also generate different fragment universes, so their F_1 values describe performance under each input condition rather than comparing identical objects. More independent fields, sealed before configuration, and multi-reader annotations are required.

One caveat from the comparisons must be separated from the claim it seems to threaten. The comparators lose on how a crossing is *resolved*, which the metric scores and which is the difficult decision, not on how it is *represented*, which the metric cannot see: GraFT's near-hard partition is a real limitation that costs it nothing here. Both properties are worth having, and Section 3.4 measures the representation directly, including where PLECTA over-marks it.

Two further limits concern the output itself. PLECTA recovers projected two-dimensional identity: shared pixels encode overlap but not which filament passes above another, so no three-dimensional topology follows. And the grouping metric evaluates ownership of centreline fragments, which is not a validation of rendered width; quantitative diameter claims require matched physical measurements across the relevant scale, contrast, and specimen ranges. Pairwise fragment scores also emphasise instances with many fragments, which is why recovery and ARI are reported alongside.

The comparisons themselves are bounded. The greedy rule is our reimplementation of a principle rather than of any particular paper, and DNAi was run outside the domain it was built for, part of its cost function reading a colour channel our masks do not have. What the results support is narrow and mechanical: pairing endpoints greedily, or committing to a pairing with no option to decline, loses identity in dense crossings. They do not support a claim that PLECTA is generally superior; that would need several released implementations run in their own regimes by their own authors.

An independent procedural benchmark is the practical next test. The `merged_lines` generator in the CNT_SEM pipeline can export overlap-aware whole-bundle references from a different implementation and morphology model, but its field-of-view preset is calibrated to a narrow coverage regime, so it must supplement rather than replace the current generator, once that calibration is fixed and its seeds separated from any configuration work. No result from it is included here. Broader validation should extend to independent real masks and other filament domains, with input fragments and scoring rules held fixed.

5 Conclusions

PLECTA reconstructs overlapping filament instances from a binary axis mask by alternating exact junction matching with gated gap matching as local arm geometry expands into chain-level evidence. The fixed method achieved common-fragment $F_1 = 0.854$ on 128 independently seeded held-out scenes from the same procedural generator, with performance declining as network density increased. On identical masks it gained +0.145 F_1 over a greedy minimum-turning-angle continuation rule, the rule shared by several published linkers, and +0.478 and +0.339 over two released implementations run from their own source; these support solving each junction exactly rather than pairing endpoints in priority order, but the released comparisons are limitations of domain and density rather than a general ranking. Development ablations identify junction chord-turn evidence, gap bridging, and junction-topology consolidation as the principal contributors, and optional image-informed rendering improves geometric presentation while remaining separate from the grouping claim.

The proof of concept on carbon-nanotube sheets joins the two pipelines: an upstream synthetic-data-driven segmenter prepares the mask, and PLECTA resolves it into instances, demonstrated on two SEM fields under manual-derived and U-Net masks without establishing population generalisation or width accuracy. The method therefore offers an inspectable, deterministic route to projected filament identity; broader claims require independent real fields, validated physical widths, further released implementations run in a matched comparison, and an external-generator benchmark with sealed seeds.

CRediT author contribution statement

[To do: Fill in each author's CRediT roles before submission, checked against actual contributions; keep the author names and order as listed above.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at https://github.com/Jorallan/PLECTA. Generated scenes, detailed per-scene records, and development diagnostics are retained locally. **[To do: Create a tagged archival release and add its DOI before submission; do not infer or preassign a DOI.]**

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